

LSU OCS 4001/7001 Fall 2026 Syllabus

Scientific Computing for Ocean, Coastal and Environmental Science OCS 4001 and 7001 002 (combined lecture and lab, 3 credit hours, 4 contact hours)

Class meets M/W 3:30-5:20 with the following tentative schedule:

3:30-3:50 coding practice and warmups
3:50-4:30 Lecture
10 min break
4:40-5:20 Lab

We will be using slack and github to chat and share material.

Instructor:

Dr. Cheryl Harrison

cherylharrison@lsu.edu (but please send me a slack message)

Office hours after class by appointment, or just send me a message on slack

Mission Statement

Data analytics and scientific programming are critical, marketable research skills in science and industry. In this class we will be developing expertise in scientific programming in the python programming language from the basics up to the advanced and cutting edge, empowering you as a researcher to make the tools you need. Students will learn how to write programs, modify others' programs, analyze and visualize data. By the end of the semester, students will complete a research project relevant to their research or field of study. For students who have a programming background, this is an opportunity to quickly go through the scaffolding and focus more on the research project and advanced programming elements of this course. I am happy to make an individual development plan with you based on your personal goals.

Learning Goals

- Develop proficiency writing code in python for scientific applications using scripts and notebooks
- Python packages and how to get them
- Logical controls: for, while and do loops, etc.
- Data types, data files, input/output including netcdf files
- Visualization: scientific plotting, movies, maps
- Data analysis: statistics, machine learning
- AI for use in programming and in numerical integration (NEW!)
- Running python on supercomputers
- Other subjects as dictated by student interest
- Develop expertise related to the students' research culminating in a research project
- Project management skills
- Scientific writing and presentation skills: proposal writing, manuscript writing, bibliography management, scientific formatting in LaTeX, research presentations

Pedagogical Style

The basic format is a period of warmup and coding practice in groups, then a period of lecture/demonstrations, followed by lab time where we will hack away together. For lab, the expectation is that you will help one another, steal code from the internet, and generally hack things together until they work. Code commenting is strongly recommended.

This is the second time this class is being offered here at LSU, though it was taught previously at UTRGV three times. The goal is that we will explore together the best way to develop proficiency and have fun learning

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scientific python. Deviations from the class schedule are to be expected, and all suggestions and requests for material to be covered will be considered. We will push ourselves, but we will enjoy the process and the rewards.

It is expected that students will make progress at different rates based on their background and abilities, and that the level of difficulty in the final project will reflect this. However, effort, showing up and putting time in is required for all students. Graduate students taking the 7001 section will be required to do more research for their final project, with a more detailed proposal, additional analysis and figures, a longer final report, and a more refined final presentation. 7001 students are also expected to complete all advanced labs. Graduate students are welcome to take the 4001-level course, and this might be more appropriate for some students, such as non-thesis master's students who are not planning to do much computation in the future, or students who just need a lighter load and have their 7000 level courses done.

There is no textbook for this course (see resources below). All materials are free and online. You will need to bring a computer to class, though we *might* have a couple of laptops you can check out if pre-arranged. You should also bring headphones for video tutorials.

Final Research Project

The big goal of this class is to learn programming for scientific applications. To this end, a good chunk of the semester will be spent working on an individual research project using scientific data analysis. Students are encouraged to do a project related to their academic research in progress, or topics they wish to explore in the future. In previous classes students have used the final project to work on methods for their theses, but basically you can do whatever you are into. It is critical here that you pick a **small, tractable** project that is easy to do in a few weeks of working 10-12 hours per week on the programming and analysis, plus time for the write up.

For this project, we are doing the scientific process in miniature. Thus, you will write a research proposal on what you plan to do, present it to the class and get feedback, do the research, then write a report in the style of an academic journal article on your results, and give a presentation to the class on your results. Previous semesters' projects are available on the respective github sites for inspiration. The biggest pitfall here is trying to do too much. You can always add more, make your minimum viable product very doable and have a stretch goal.

Research project proposal:

Write a short (1-3 pages for 4001, 2-4 pages for 7001 students) research proposal, including: abstract, background, why it's important, what your hypothesis is, what analysis you plan to do, what the broader impacts are, and a timeline for completion. (Broader impacts: How can this information be used? Why do we care? Impact for society...) At least one figure and three references are required for 4001 students, and two figures and five references for 7001 students. You will also give a short class presentation to get feedback on your proposed research and presenting skills.

Research report:

5-10 pages for 4001 students, 8-12 pages for 7001 students. Should take the form of a scientific article with the following sections: Abstract, Introduction, Methods, Results, Discussion, Bibliography. 2-5 figures or tables (4-7 for 7001) and bibliography are required.

Final presentation using powerpoint or your favorite presentation software.

Detailed rubrics with a checklist of expectations for satisfactory performance will be provided for each component.

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Grading

Letter grades will be assigned based on attendance, labs, the final project and course feedback. Assignments will be submitted on the class github and the class Moodle site. Labs are due Sunday nights, but it's best to have them done before the next class so the work doesn't pile up.

We are using [specification grading](#), which means all assignments will be graded satisfactory or unsatisfactory, and specified groups of assignments need to be completed to receive a certain grade. Rubrics for satisfactory completion of assignments will be outlined when the assignment is given, but in general the assignment must be complete and of appropriate length and sufficient content. Assignments will be graded as satisfactory or unsatisfactory, and you will need to resubmit unsatisfactory labs in a timely manner.

Since programming, like mathematics, builds on previous lessons, skipping assignments will be problematic and will result in a very poor grade in the course. Note if you miss a class meeting you are still responsible for all the material. Specified assignments for receiving a D grade cannot be skipped, and their satisfactory and timely completion is required to pass the course. Course feedback is through weekly assessments on the class and on student's performance, and a class and self-assessment at the end of the term. The design philosophy of this course is that it will adapt to students' needs throughout the course.

To receive an A grade, students must

- Meet the specifications for receiving a B grade
- Satisfactorily complete **three (four for 7001)** advanced labs (Labs 12.2-14.1)

To receive a B grade, students must

- Meet the specifications for receiving a C grade
- Satisfactorily complete **two (three for 7001)** advanced labs (Labs 12.2-14.1)

To pass the course with a C, students must

- Meet the specifications for receiving a D grade
- Satisfactorily complete the Statistics and Timeseries Analysis labs (Labs 11.1-12.1)
- Provide satisfactory peer feedback

To pass the course with a D, students must

- Satisfactorily complete all assigned chapters in datacamp and the associated labs, as well as Cartopy and Movies labs (Labs 1.1 – 10.2)
- Satisfactorily complete all components of the final project

Failure to deliver a satisfactory final project will result in an F, regardless of other assignments done. For +/- modifications on the specified grades, performance on the final project will be considered, along with attendance and submission of weekly assessment forms.

If you have an issue come up, i.e. you get sick or have some other unfortunate circumstance, have fieldwork, etc., please communicate with me sooner rather than later and I will do everything in my power to give you reasonable accommodation and help you out. Life is inevitably complicated, and we all need to work together to support one another.

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Resources

All resources for this class are free and online

AN – Anaconda python/R platform

<https://www.anaconda.com/download>

C – Cartopy, Python mapping plotting package

<https://scitools.org.uk/cartopy/docs/latest/>

DC- datacamp, see assignments for class datacamp login info

Datacamp has interactive python (and other language) lessons. We get free access to this for our course.

O – Overleaf, Scientific formatting software using the LaTeX markup language

<https://www.overleaf.com/>

Introduction to LaTeX:

https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes

P – PANDAS (Python Data Analysis Library)

<https://pandas.pydata.org/>

R – The R programming language

<https://swirlstats.com/> is a great intro tutorial, and tells you how to install R

<https://r4ds.hadley.nz/> data science in R (more advanced)

SP – Scientific Python

Robert Johansson's scientific python lectures:

<https://github.com/jrjohansson/scientific-python-lectures>

Scipy lecture notes

<http://scipy-lectures.org/>

Matplotlib (plotting package)

<https://matplotlib.org/>

Scipy documentation (more advanced)

<https://docs.scipy.org/doc/scipy/tutorial/index.html>

W - On Scientific Writing:

https://cbc.arizona.edu/sites/default/files/2023-08/Scientific_Writing.pdf

<https://writingcenter.unc.edu/tips-and-tools/sciences/>

<https://advice.writing.utoronto.ca/types-of-writing/science/>

<https://github.com/quanghuy0497/Writing-in-the-Sciences>

XR – XARRAY, the array package for python

<http://xarray.pydata.org/en/stable/>

Z – Zotero, Bibliography software

<https://www.zotero.org/>

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Tentative Class Schedule. Advanced labs are highlighted. Major deadlines are in yellow. Holidays are in green. Weekly labs and assessments (WA) are due Sunday at midnight in Moodle and github.

Week	Dates	Topic	Resources (defined above)	Assignments
1	Aug 24, 26	Introductions: class, python	DC Intro to Python Ch 1-2	Lab 1.1, Lab 1.2, Weekly assessment (WA), Student Introductions
2	Aug 31, Sep 2	Functions, Numpy	DC Intro to Python Ch 3-4	Lab 2.1, Lab 2.2, WA
3	Labor Day Holiday, Sep 9	Matplotlib, Pandas	DC Intermediate Python Ch 1-2	Lab 3.1, WA
4	Sep 14, 16	Logical control, Loops, Show us your plots	DC Intermediate python Ch 3-4	Lab 4.1, Lab 4.2, WA
5	Sep 21, 23	Case Study, Functions	DC Intermediate python Ch 5, DC Intro to Functions Ch 1	Lab 5.1, Lab 5.2, WA
6	Sep 28, 30	Plotting: More matplotlib, Xarray, Cartopy	DC Introduction to Data Visualization with Matplotlib	Lab 6.1, Lab 6.2, WA
7	Oct 5, 7	More Pandas, Movies	DC Data Manipulation with Pandas	Final Research topics due 10/11 Lab 7.1, Lab 7.2, WA
8	Oct 12, 14	LLMs and Programming, Movie share	O, Z, R, S	Lab 8.1 WA
9	Oct 19, 21	Pseudocode, Introduction to LaTeX and Zotero, LSU HPC, Research projects		Final project proposals due 10/25 Lab 9.1, Lab 9.2, WA
10	Oct 26, 28	Research topics presentations, Proposal peer review		Research topics feedback, Lab 10.2, WA
11	Nov 2, 4	Statistics	DC Introduction to Statistics in Python	Lab 11.1, Lab 11.2, WA
12	Nov 9, 11	Timeseries Analysis, Predator Prey Modeling		Lab 12.1, Lab 12.2, WA
13	Nov 16, 18	Intro to Machine Learning	DC Supervised Learning with scikit-learn	Lab 13.1, Lab 13.2, WA
14	Nov 23, Thanksgiving	AI in Numerical Modeling		Lab 14.1, WA
15	Nov 30, Dec 2	Research Projects		Research notebook, WA
16	Friday, Dec 11	Research presentations	During Final Period: Dec 11 12:30-2:30	Final paper due 12/10